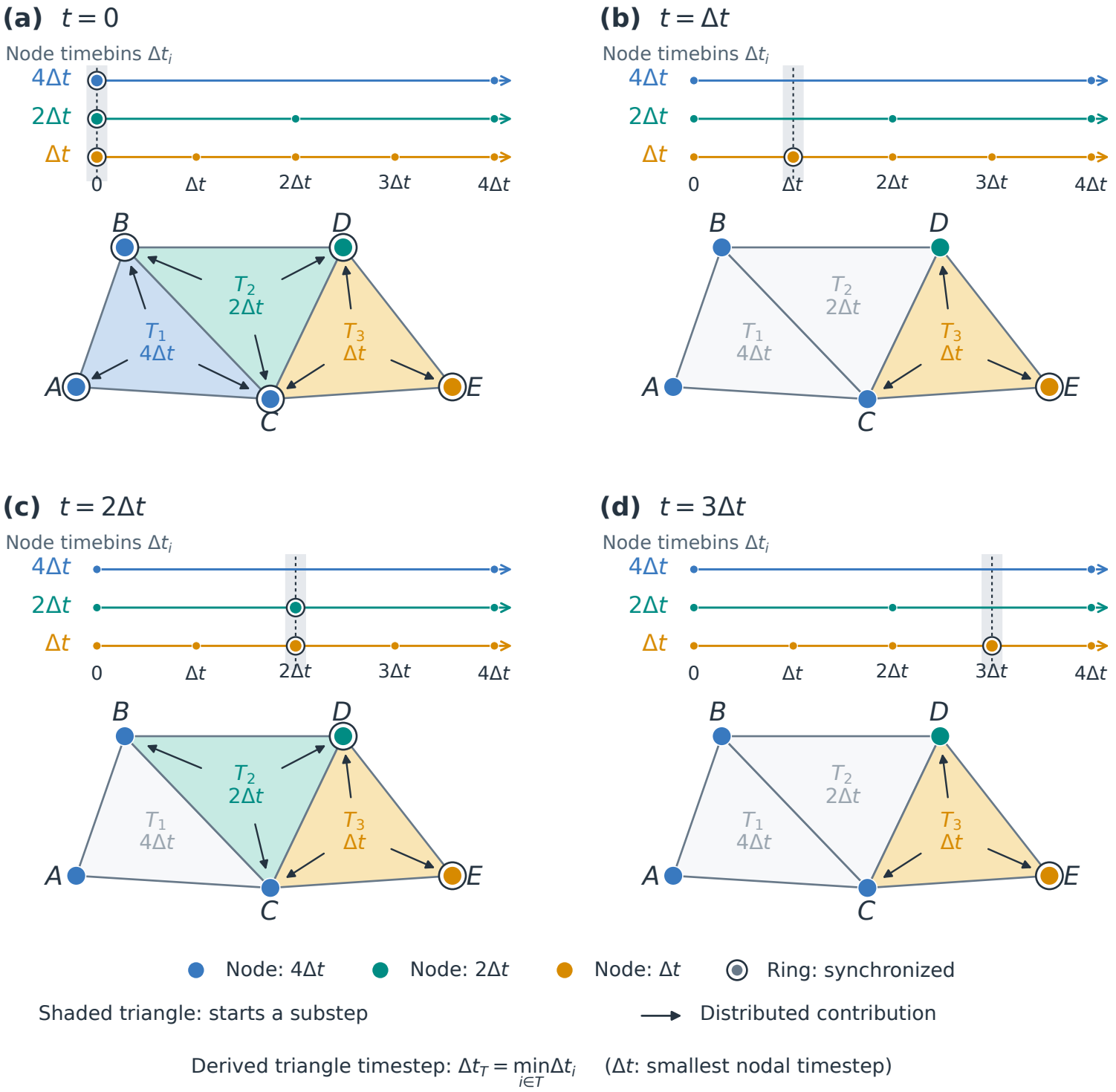
(a) $t = 0$
Node timebins $\Delta t_i$
$4\Delta t$
$2\Delta t$
$\Delta t$
0
$\Delta t$
$2\Delta t$
$3\Delta t$
$4\Delta t$
B
D
$T_2$
$2\Delta t$
$T_1$
$4\Delta t$
$T_3$
$\Delta t$
A
C
E
(b) $t = \Delta t$
Node timebins $\Delta t_i$
$4\Delta t$
$2\Delta t$
$\Delta t$
0
$\Delta t$
$2\Delta t$
$3\Delta t$
$4\Delta t$
B
D
$T_2$
$2\Delta t$
$T_1$
$4\Delta t$
$T_3$
$\Delta t$
A
C
E
(c) $t = 2\Delta t$
Node timebins $\Delta t_i$
$4\Delta t$
$2\Delta t$
$\Delta t$
0
$\Delta t$
$2\Delta t$
$3\Delta t$
$4\Delta t$
B
D
$T_2$
$2\Delta t$
$T_1$
$4\Delta t$
$T_3$
$\Delta t$
A
C
E
(d) $t = 3\Delta t$
Node timebins $\Delta t_i$
$4\Delta t$
$2\Delta t$
$\Delta t$
0
$\Delta t$
$2\Delta t$
$3\Delta t$
$4\Delta t$
B
D
$T_2$
$2\Delta t$
$T_1$
$4\Delta t$
$T_3$
$\Delta t$
A
C
E
Node: $4\Delta t$
Node: $2\Delta t$
Node: $\Delta t$
Ring: synchronized
Shaded triangle: starts a substep
Distributed contribution
Derived triangle timestep: $\Delta t_T = \min_{i \in T} \Delta t_i$ ($\Delta t$: smallest nodal timestep)